

Disproof of Conjecture 000000001066

Exact square-root attainment over prime fields

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Abstract

The conjecture asserts a cardinality bound for Sidon subsets of $\mathbb{F}_p$ and attainment of $\sqrt{p}$ for prime p . We disprove the exact-attainment clause for every prime: a finite set has integer cardinality, whereas the square root of a prime is not an integer. The accompanying Lean project proves the negation of the full written conjunction under both universal-prime and existential-prime readings of the attainment clause.

1 Statement and interpretation

The supplied English conjecture states:

Definition: A Sidon set $B \subset \mathbb{F}_p$ (all pairwise differences distinct). Conjecture: $|B| \leq p^{1/2} + 1$, and $p^{1/2}$ is attained for prime p (finite-field Sidon extremality).

The Chinese version has the same bound and attainment clause, joined by “and.” The exact, unmodified bilingual source is included as `conjecture.md`.

We use the conventional difference formulation of the Sidon property: the map $(a, b) \mapsto a - b$ is injective on

$$D_B = \{(a, b) \in B \times B : a \neq b\}.$$

Thus differences are ordered and diagonal pairs are excluded. The symbol $p^{1/2}$ denotes the nonnegative real square root, with the natural numbers p and $|B|$ embedded in $\mathbb{R}$. “Attained” is read as the equality $|B| = \sqrt{p}$. Neither language specifies rounding or an asymptotic qualifier.

Write $U(p)$ for the stated bound for every Sidon subset of $\mathbb{F}_p$, and $A(p)$ for the existence of a Sidon subset of cardinality exactly $\sqrt{p}$. The standard reading is

$$C_{\forall} : \quad \forall p \text{ prime}, \quad U(p) \wedge A(p).$$

For completeness, we also treat the weaker prime quantifier

$$C_{\exists} : \quad (\forall p \text{ prime}, U(p)) \wedge (\exists p \text{ prime}, A(p)).$$

The argument below rules out $A(p)$ for every prime, so it refutes both readings without any claim about the truth of $U(p)$.

2 The cardinality obstruction

Theorem 1. *For every prime p and every finite subset $B \subseteq \mathbb{F}_p$, one has $|B| \neq \sqrt{p}$.*

Proof. Put $n = |B| \in \mathbb{N}$ and suppose $n = \sqrt{p}$ as real numbers. Since $p \geq 0$, squaring gives $n^2 = p$; this is also an equality of natural numbers by injectivity of their embedding in the reals. If $n = 0$ or $n = 1$, then $p = 0$ or $p = 1$, contradicting primality. If $n \geq 2$, then $p = n \cdot n$ has a proper divisor n , since $1 < n < n^2 = p$, again contradicting primality. $\square$

The theorem applies to all finite subsets, hence to Sidon subsets. It gives $\neg A(p)$ and consequently $\neg(U(p) \wedge A(p))$ for every prime p . Specializing to the prime 2 yields $\neg C_{\forall}$. Universal failure of $A(p)$ also yields $\neg C_{\exists}$.

3 Formalization and verification

The project uses Lean 4.19.0 and Mathlib v4.19.0, with exact dependency revisions pinned in `lean/lake-manifest.json`. Its objects are actual finite subsets `Finset (ZMod p)`; for prime p , `ZMod p` is the prime field. All such subsets are finite, so this representation covers the subsets in the conjecture.

The namespace `Conjecture1066` contains five definitions:

```
IsSidon, UpperBound, ExactAttainment,
UniversalConjecture, ExistentialPrimeConjecture.
```

The central theorem is `nat_cast_ne_sqrt_prime`. It uses `Real.sq_sqrt` and exact transfer of the natural-number equality from the reals. The prime-power obstruction is `Nat.Prime.not_prime_pow`, used at exponent two.

The final theorems are

```
universal_conjecture_false    and
existential_prime_conjecture_false.
```

A fresh copied project was built successfully. Both the proof source and `Check.lean` were replayed with warnings treated as errors. The audit prints all five definitions and all seven theorem types and axiom dependencies, and separately inventories all twelve constants originating in the compiled proof module. Every theorem depends only on `propext`, `Classical.choice`, and `Quot.sound`. There are no admitted proofs, custom axioms, or `native_decide` calls. The proof uses no auxiliary search or numerical computation; its sole decidable numeral check certifies $2 \leq 2$ with ordinary kernel-checked `decide`.

Reproduction instructions, complete execution and trust-audit records, and file hashes accompany the project. These are local validation records; repository acceptance is a separate maintainer decision.

4 Scope of the disproof

The submission refutes literal exact attainment of $\sqrt{p}$ over prime fields, hence the written conjunction. It does not prove or refute the upper-bound clause alone, determine a largest Sidon set, or address rounded or asymptotic attainment. Replacing the prime parameter by a prime power, adding a floor, or changing exact equality to an asymptotic statement would produce different claims. None of those changes is made here.